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Chapter 2

Introduction to System Modeling

2.1 Introduction

A **mathematical model** is an equation or set of equations which adequately describes the behavior of a system.

System modeling is a subject in its own right. Essentially there are two approaches to finding the model, (Dutton et al., 1997).

- In the first, the system is broken down into smaller elements. For each element a mathematical description is then established by working from the physical laws which describe the system's behavior. The simplest such technique is lumped-parameter modeling, which is considered in this course.
- The second approach is known as system identification in which it is assumed that an experiment can be carried out on the system, and that a mathematical model can be found from the results. This approach can clearly only be applied to existing plants, whereas lumped-parameter modeling can additionally be applied to a plant yet to be built, working purely from the physics of the proposed plant components.

System modeling is a specialization of the more general area of mathematical modeling. In a system model, the important relationship is that between the manipulated inputs and measurable outputs. Ideally this relationship should be linear and capable of being described by an expression of low-order (that is, an equation or set of equations containing as few terms as possible). An usual graphical representation of single input, single output (SISO) dynamical system is shown in Figure 2.1. A model of such a system is a mathematical relationship between the input $u(t)$ and the output $y(t)$.

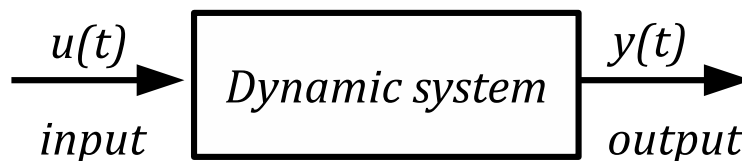


Figure 2.1: SISO dynamic system

Most commonly, a low-order linear differential equation model is used.

When fundamental physical laws are applied to the lumped-parameter model, the resulting equations may be nonlinear in which case further assumptions may have to be made in order to produce an ordinary linear differential equation model which is soluble. In such cases, it is not unusual to assume that system operation will be restricted to small perturbations about a given operating point. If the assumed operating region is small enough, most nonlinear plants may be described by a set of linear equations. If these assumptions cannot be made, nonlinear control techniques are needed.

2.2 Lumped-parameter models

The systems studied in this course are:

Linear - They must obey the principle of superposition that is, if an input $u_1(t)$ causes an output $y_1(t)$ and an input $u_2(t)$ causes an output $y_2(t)$, then an input $u_1(t) + u_2(t)$ causes an output $y_1(t) + y_2(t)$ (see next section **Linear Approximations of Physical systems**)

In the case of the resistor, if an input of 1A produces an output of 3V, and an input of 2A produces an output of 6V then an input of 3A produces an output of 9V.

Stationary (or time invariant) - The parameters inside the element must not vary with time. In other words, an input applied today must give the same result as the same input applied yesterday or tomorrow.

A vehicle which burns large masses of fuel, such as racing cars or space vehicles, is an example of system which is **not** stationary (or it is time-varying). Its dynamic behavior will alter significantly as its mass decreases, (Dutton et al., 1997).

Deterministic - The outputs of the system at any time can be determined from a knowledge of the system's inputs up to that time. In other words there is no random (or stochastic) behavior in the system, since its outputs are always a specific function of the inputs. The term *causal* is also used for such systems, (Dutton et al., 1997).

Example 2.2.1 Consider an element representing an idealized component such as:

- The resistor. If the resistor is linear then its model is a relationship between the voltage $v(t)$ and the current $i(t)$:

$$i(t) = \frac{1}{R}v(t)$$

- The inductor is described by the equation:

$$i(t) = \frac{1}{L} \int v(t)dt \quad \text{or} \quad v(t) = L \frac{di(t)}{dt}$$

- The capacitor is described by:

$$i(t) = C \frac{dv(t)}{dt}$$

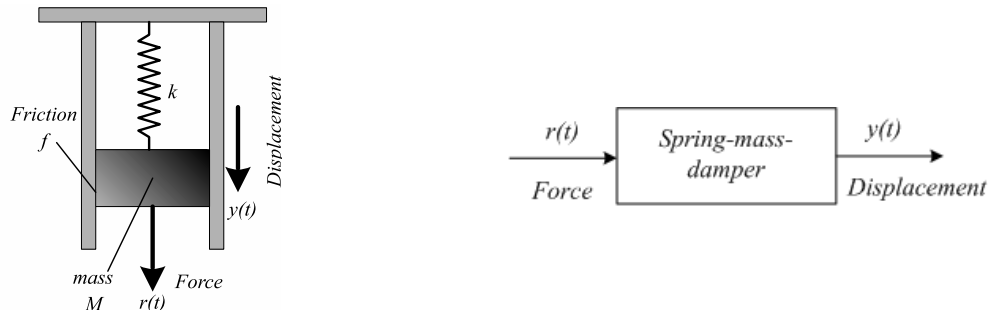


Figure 2.2: Spring-mass-damper system

- The spring-mass-damper mechanical system (Figure 2.2). The system could represent for example an automobile shock absorber. We model the wall friction as a viscous damper. Assuming the system is linear, a model of the system can be derived from the Newton's second law of motion.

The force equation of this system is:

$$M \frac{d^2 y(t)}{dt^2} = -f \frac{dy(t)}{dt} - ky(t) + r(t)$$

where f is the friction coefficient, M - the mass, k - the stiffness of the linear spring. The derivatives $\frac{d^2 y(t)}{dt^2}$ and $\frac{dy(t)}{dt}$ represent the acceleration and velocity of the mass, respectively.

Alternatively, the former equation may be rewritten into an input-output form:

$$M \frac{d^2 y(t)}{dt^2} + f \frac{dy(t)}{dt} + ky(t) = r(t)$$

It is a relationship between the (input) force $r(t)$ and the (output) displacement $y(t)$.

2.3 Linear Approximation of Physical Systems

A great majority of physical systems are linear within some domain of variables. However, all systems ultimately become nonlinear as the variables are increased without limit.

A system is defined as linear in terms of the system variables. In the case of an electrical network, the excitation is the input current and the response is the voltage. In general a *necessary condition* for a linear system can be determined in terms of an excitation $x(t)$ and a response $y(t)$. When a system at rest is subjected to an excitation $x_1(t)$, it provides a response $y_1(t)$. Furthermore, when the system is subjected to an excitation $x_2(t)$, it provides a corresponding response $y_2(t)$. For a linear system, it is *necessary* that the excitation $x_1(t) + x_2(t)$ result in a response $y_1(t) + y_2(t)$. This is usually called the *principle of superposition*.

Furthermore, it is necessary that the magnitude scale factor be preserved in a linear system. Again, consider a system with an input x and an output y . Then it is necessary that the response of a linear system to a constant multiple β of an input x be equal to the response to the input multiplied by the same constant so that the output is equal to βy . This is called the property of *homogeneity*.

Example 2.3.1 , (Dorf and Bishop, 2008). A system characterized by the relation $y = x^2$ is not linear because the superposition property is not satisfied. A system represented by the relation

$y = mx + b$ is not linear because it does not satisfy the homogeneity property. However, this device may be considered linear about an operating point x_0, y_0 for small changes Δx and Δy . When $x = x_0 + \Delta x$ and $y = y_0 + \Delta y$, we have

$$y = mx + b$$

or

$$y_0 + \Delta y = mx_0 + m\Delta x + b$$

and because

$$y_0 = mx_0 + b$$

then:

$$\Delta y = m\Delta x$$

which satisfies the necessary conditions (Figure 2.3).

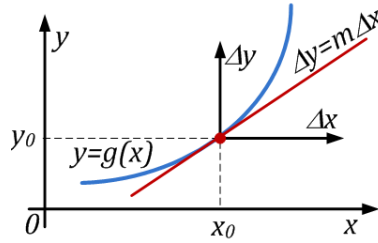


Figure 2.3: Linear approximation

Consider a general element with an input $x(t)$ and a response $y(t)$. The relationship of the two variables is written as:

$$y(t) = g(x(t))$$

where $g(x(t))$ is a continuous function of $x(t)$.

In general, Taylor series may be used to expand a nonlinear function $g(x(t))$ about an operating value x_0 . An operating value could be the equilibrium position in a spring-mass-damper, a fixed voltage in an electrical system, steady-state pressure in a fluid system, and so on.

Because the function is continuous over the range of interest, a Taylor series expansion about the operating point x_0 may be utilized. Then we have:

$$y = g(x) = g(x_0) + \frac{dg}{dx}\bigg|_{x=x_0} \frac{x - x_0}{1!} + \text{higher order terms} \quad (2.1)$$

The slope at the operating point,

$$m = \frac{dg}{dx}\bigg|_{x=x_0},$$

is a good approximation to the curve over a small range of $(x - x_0)$, the deviation from the operating point. Then, as a reasonable approximation, the expression from example 1 becomes:

$$y = g(x_0) + \frac{dg}{dx}\bigg|_{x=x_0} (x - x_0) = y_0 + m(x - x_0),$$

where m is the slope at the operating point. Finally, this equation can be rewritten as the linear equation

$$(y - y_0) = m(x - x_0)$$

or

$$\Delta y = m\Delta x.$$

Example 2.3.2 *Pendulum oscillator model, (Dorf and Bishop, 2008).*

Consider the pendulum oscillator shown in Figure 2.4. The torque on the mass is

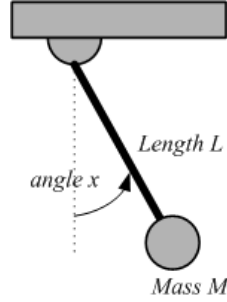


Figure 2.4: Pendulum oscillator

$$T = MgL\sin(x)$$

where g is the gravity constant. The equilibrium condition for the mass is $x_0 = 0^\circ$. The first derivative evaluated at equilibrium provides the linear approximation, which is:

$$T - T_0 \cong MgL \frac{\partial \sin x}{\partial x} \Big|_{x=x_0} (x - x_0),$$

where $T_0 = 0$. Then, we have

$$T = MgL(\cos 0^\circ)(x - 0^\circ) = MgLx$$

The approximation is reasonably accurate for $-\pi/4 \leq x \leq \pi/4$. For example, the response of the linear model for the swing through $\pm 30^\circ$ is within 2% of the actual nonlinear pendulum response (check in simulations).

If the variable y depends upon several excitation variables $x_1, x_2, \dots, x_n$, then the functional relationship is written as:

$$y = g(x_1, x_2, \dots, x_n).$$

The Taylor series expansion about the operating point $x_{10}, x_{20}, \dots, x_{n0}$ is useful for a linear approximation to a nonlinear function or a nonlinear differential equation. When the higher-order terms are neglected, the linear approximation is written as

$$y = g(x_{10}, x_{20}, \dots, x_{n0}) + \frac{\partial g}{\partial x_1} \Big|_{x=x_0} (x_1 - x_{10}) + \frac{\partial g}{\partial x_2} \Big|_{x=x_0} (x_2 - x_{20}) + \dots + \frac{\partial g}{\partial x_n} \Big|_{x=x_0} (x_n - x_{n0}) \quad (2.2)$$

where $x_0 = [x_{10}, x_{20}, \dots, x_{n0}]$ is the operating point. An example will illustrate this method.

Example 2.3.3 Magnetic levitation, (Xie, 2009).

The device shown in Figure 2.5 is an actively controlled magnetic levitator. Magnetic levitation is a technique widely used to create noncontact bearings, to eliminate friction for operation at high speeds or in special environments such as vacuum systems. In a simplified approach, the system consists of an iron-core electromagnet and the steel ball levitated by the electromagnet.

Current through the coils of the electromagnet induces a magnetic field in the iron core. The electromagnet exerts a force on the steel ball so as to keep it floating at a given height (in conjunction with a sensor and control electronics).

A mathematical model for this system can be derived from basic physics. As shown in the free body diagram (Figure 2.5), two forces act on the steel ball: gravity and the electro-magnetic force F_m . In the simplest model, F_m is proportional to the square of the current passing through the inductor and inversely proportional to the square of the distance between the magnet and the steel ball:

$$F_m = C \frac{i^2(t)}{z^2(t)}$$

where the constant C depends on the physical construction of the electromagnet.

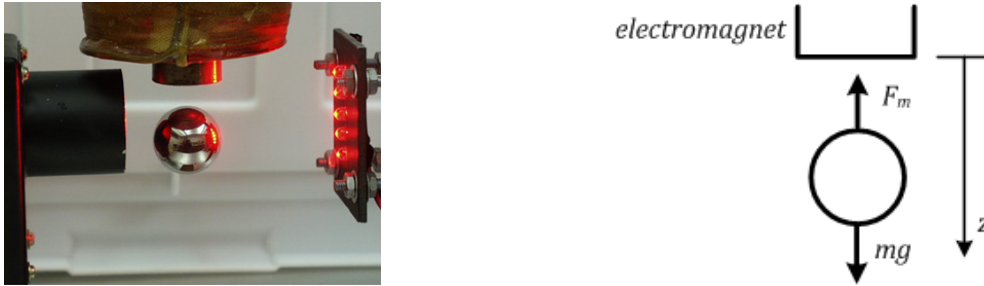


Figure 2.5: Magnetic levitation and the free body diagram

We will determine a nonlinear model for this system and a linear approximation when the ball is in equilibrium for a distance $z_0 = 10^{-2}m$ from the electromagnet. Consider the following values for the constants: the gravitational acceleration $g = 10m/s^2$, mass of the ball $m = 0.1kg$ and the constant $C = 10^{-2}Nm^2/A^2$.

The equation of motion for the ball is:

$$m\ddot{z}(t) = mg - F_m$$

or

$$m\ddot{z}(t) = mg - C \frac{i^2(t)}{z^2(t)}$$

where $\ddot{z}(t) = \frac{d^2z(t)}{dt^2}$. From the description of this system, the input signal (the cause) is the current through the coils of the electromagnet $i(t)$ and the output (the effect) is the displacement of the ball $z(t)$ (Figure 2.6).

The differential equation obtained above is a nonlinear relationship between $i(t)$ and $z(t)$, thus it is a mathematical model for this system.

The equation can also be written as a function of 3 variables: the acceleration $\ddot{z}(t)$, the

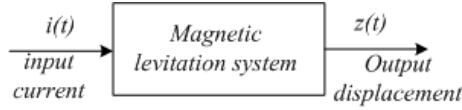


Figure 2.6: Magnetic levitation system. Input and output

position $z(t)$ and the current $i(t)$, that equals zero:

$$g(\ddot{z}(t), z(t), i(t)) = m\ddot{z}(t) - mg + C \frac{i^2(t)}{z^2(t)} = 0 \quad (2.3)$$

The first step in linearizing this equation is setting the desired operating point. If the ball is at equilibrium for z_0 , then the speed and acceleration of the ball are zero for this distance: $\ddot{z}_0 = 0$. The current that keeps the ball floating at the distance z_0 is obtained from the nonlinear equation for $z(t) = z_0$ and $\ddot{z}(t) = \ddot{z}_0$:

$$m\ddot{z}_0 = mg - C \frac{i_0^2}{z_0^2}$$

or

$$0 = mg - C \frac{i_0^2}{z_0^2}, \Rightarrow i_0 = \sqrt{\frac{mg}{C}} z_0 = 0.1A$$

The operating point for linearization is then: $(\ddot{z}_0, z_0, i_0) = (0, 0.01, 0.1)$

According to (2.2), the Taylor series for the approximation of the nonlinear function (2.3) is written as

$$\begin{aligned} 0 = g(\ddot{z}(t), z(t), i(t)) &\approx g(\ddot{z}_0, z_0, i_0) + \frac{\partial g}{\partial \ddot{z}}|_{(\ddot{z}_0, z_0, i_0)} \cdot (\ddot{z}(t) - \ddot{z}_0) + \\ &+ \frac{\partial g}{\partial z}|_{(\ddot{z}_0, z_0, i_0)} \cdot (z(t) - z_0) + \frac{\partial g}{\partial i}|_{(\ddot{z}_0, z_0, i_0)} \cdot (i(t) - i_0) \end{aligned}$$

or:

$$0 \approx 0 + m \cdot (\ddot{z}(t) - \ddot{z}_0) - 2C \frac{i_0^2}{z_0^3} \cdot (z(t) - z_0) + 2C \frac{i_0}{z_0^2} \cdot (i(t) - i_0)$$

We denote by $\Delta\ddot{z}(t) = \ddot{z}(t) - \ddot{z}_0$, $\Delta z(t) = z(t) - z_0$ and $\Delta i(t) = i(t) - i_0$ the variations of variables around the operating point, and re-arranging the equation above, we obtain:

$$m\Delta\ddot{z}(t) = 2C \frac{i_0^2}{z_0^3} \Delta z(t) - 2C \frac{i_0}{z_0^2} \Delta i(t) \quad (2.4)$$

All coefficients are constant and the relation (2.4) is a linear and homogeneous differential equation. By replacing the constant values, the equation (2.4) is:

$$0.1\Delta\ddot{z}(t) = 200\Delta z(t) - 20\Delta i(t)$$

2.4 The Laplace Transform

The ability to obtain linear approximations of physical systems allows the analyst to consider the use of Laplace transformation. The Laplace transform method substitutes the more difficult

differential equations with the relatively easily solved algebraic equations. The basic Laplace transform of a time signal $f(t)$ is defined as:

$$F(s) = \int_0^{\infty} f(t)e^{-st} dt$$

and written symbolically as: $F(s) \equiv \mathcal{L}[f(t)]$

It is common practice to use a capital letter F which is a function of the new variable s for the transform of the time signal $f(t)$. Also it is assumed that $f(t)$ is zero for all times before $t=0$. In the definition of Laplace transform the exponent st must be dimensionless otherwise the expression e^{-st} is meaningless. Thus the variable s has dimension of 1/time which is the dimension of frequency. Since s is also a complex quantity it is often referred to as the *complex frequency*.

Table 2.1: Laplace transform operations

1	Transform integral	$f(t)$	$\int_0^\infty f(t)e^{-st}dt$ or $\mathcal{L}[f(t)]$
2	Linearity	$f_1(t) \pm f_2(t)$	$F_1(s) \pm F_2(s)$
3	Constant multiplication	$af(t)$	$aF(s)$
4	Complex shift theorem	$e^{\pm at}f(t)$	$F(s \pm a)$
5	Real shift theorem	$f(t-T)$	$e^{-Ts}F(s)$, $T \geq 0$
6	Scaling theorem	$f(\frac{t}{a})$	$aF(as)$
7	First derivative	$\frac{d}{dt}f(t)$	$sF(s) - f(0)$
8	n-th derivative	$\frac{d^n}{dt^n}f(t)$	$s^n F(s) - \sum_{r=1}^n \frac{d^{r-1}}{dt^{r-1}}f(0)s^{n-r}$
9	First integral	$\int_0^t f(t)dt$	$\frac{1}{s}F(s)$

Table 2.2: Laplace transforms of common functions

	$f(t)$	$F(s)$
1	Unit impulse (Dirac) $\delta(t)$	1
2	Unit step $u(t)=1$	$\frac{1}{s}$
3	Unit ramp $v(t)=t$	$\frac{1}{s^2}$
4	e^{at}	$\frac{1}{s-a}$
5	$\cos \omega t$	$\frac{s}{s^2 + \omega^2}$
6	$\sin \omega t$	$\frac{\omega}{s^2 + \omega^2}$

2.5 Signals

A few commonly used signals will be introduced in this section:

1. The unit step is defined by (Figure 2.7):

$$u(t) = \begin{cases} 0, & t < 0 \\ 1, & t \geq 0 \end{cases} \quad (2.5)$$

The Laplace transform of the step function, according to Table 2.2, is:

$$\mathcal{L}[u(t)] = \frac{1}{s}, \quad (2.6)$$

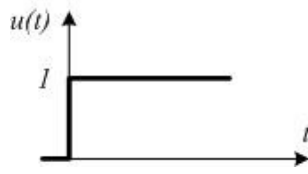


Figure 2.7: Unit step signal

2. The **unit ramp** is shown in Figure 2.8. It is defined by:

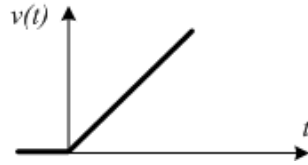


Figure 2.8: The ramp signal

$$v(t) = \begin{cases} 0, & t < 0 \\ t, & t \geq 0 \end{cases} \quad (2.7)$$

The Laplace transform of the ramp signal is:

$$\mathcal{L}[v(t)] = \frac{1}{s^2} \quad (2.8)$$

3. The **ideal impulse** also called Dirac impulse is shown in Figure 2.9 It is defined by:

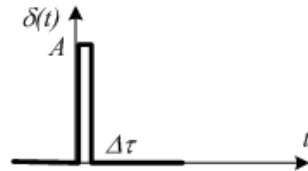


Figure 2.9: The impulse signal

$$\delta(t) = \begin{cases} 0, & t < 0 \text{ and } t > \Delta\tau \\ A, & 0 \leq t \leq \Delta\tau \end{cases} \quad (2.9)$$

where ideally

$$\lim_{\Delta\tau \rightarrow 0} \int_0^{\Delta\tau} \delta(t) dt = 1 \text{ and } A \rightarrow \infty \quad (2.10)$$

The Dirac impulse is a signal with unit area and infinite amplitude.

The Laplace transform of the unit impulse is:

$$\mathcal{L}[\delta(t)] = 1 \quad (2.11)$$

2.6 Input/output models

2.6.1 The transfer function

The **transfer function** of a linear system is defined as the ratio of the Laplace transform of the output variable to the Laplace transform of the input variable, with all the initial conditions assumed to be zero.

A transfer function may be defined only for a linear stationary system. A transfer function is an input-output description of the behavior of a system.

A linear differential equation that describes such a system can be:

$$a_0 r(t) + a_1 \frac{dr(t)}{dt} + \dots + a_m \frac{d^m r(t)}{dt^m} = b_0 y(t) + b_1 \frac{dy(t)}{dt} + \dots + b_n \frac{d^n y(t)}{dt^n}$$

where $r(t)$ and $y(t)$ are the input and output variables, as shown in Figure 2.10 where $R(s) = \mathcal{L}[r(t)]$ and $Y(s) = \mathcal{L}[y(t)]$.

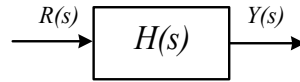


Figure 2.10: Block diagram of a system

Applying the Laplace transform for the initial conditions zero, the differential equation becomes:

$$(a_0 + a_1 s + \dots + a_m s^m)R(s) = (b_0 + b_1 s + \dots + b_n s^n)Y(s)$$

and the transfer function will then be:

$$H(s) = \frac{Y(s)}{R(s)} = \frac{a_0 + a_1 s + \dots + a_m s^m}{b_0 + b_1 s + \dots + b_n s^n}$$

Example 2.6.1 Spring-mass-damper

The transfer function of the spring-mass-damper system, shown in Figure 2.2, is obtained from the original equation model by applying the Laplace transform:

$$M \frac{d^2 y(t)}{dt^2} + f \frac{dy(t)}{dt} + ky(t) = r(t)$$

$$Ms^2 Y(s) + f s Y(s) + k Y(s) = R(s)$$

$$H(s) = \frac{Y(s)}{R(s)} = \frac{1}{Ms^2 + fs + k}$$

The block diagram now contains all the information given in the transfer function model, that is, "output = contents $\times$ input". $H(s)$ is called a transfer function. It shows how the input is transferred to the output.

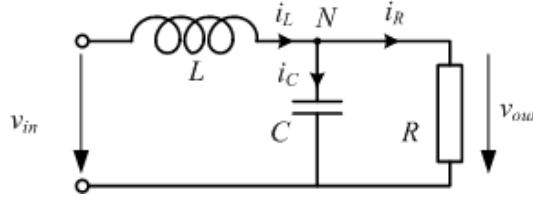


Figure 2.11: Electrical system

Example 2.6.2 Electrical system. *The transfer function for the system shown in Figure 2.11 can be derived as follows: Write the fundamental relation for each element:*

$$\text{Inductor: } \frac{di_L}{dt} = \frac{1}{L}v_L \quad (2.12)$$

$$\text{Capacitor: } \frac{dv_C}{dt} = \frac{1}{C}i_C \quad (2.13)$$

$$\text{Resistor: } v_R = Ri_R \quad (2.14)$$

Write the Kirchhoff's current law at node N:

$$i_L = i_C + i_R \quad (2.15)$$

Write the Kirchhoff's voltage law:

$$v_{in} = v_L + v_C \quad (2.16)$$

$$v_C = v_R = v_{out} \quad (2.17)$$

Assume the initial conditions zero, replace (2.17) and apply the Laplace transform to the relations (2.12) - (2.16):

$$sI_L(s) = \frac{1}{L}V_L(s) \quad (2.18)$$

$$sV_{out}(s) = \frac{1}{C}I_C(s) \quad (2.19)$$

$$V_{out}(s) = RI_R(s) \quad (2.20)$$

$$I_L(s) = I_C(s) + I_R(s) \quad (2.21)$$

$$V_{in}(s) = V_L(s) + V_{out}(s) \quad (2.22)$$

Calculate to eliminate I_L , I_C , I_R , V_L from (2.18) - (2.22):

$$V_{in} = sLI_L + V_{out} = sL(I_C + I_R) + V_{out} = sL(sCV_{out} + \frac{1}{R}V_{out}) + V_{out} = (LCs^2 + \frac{L}{R}s + 1)V_{out} \quad (2.23)$$

The transfer function is:

$$H(s) = \frac{V_{out}(s)}{V_{in}(s)} = \frac{1}{LCs^2 + \frac{L}{R}s + 1} = \frac{R}{RLCs^2 + Ls + R} \quad (2.24)$$

For a **real physical** system the function $H(s)$ is a ratio of two polynomials such that

$$H(s) = \frac{N(s)}{D(s)}$$

with the degree of the denominator $D(s)$ being greater than or equal to the degree of the numerator polynomial $N(s)$. Such a system is called **proper**. In a proper system, the **system order** is the degree of the denominator polynomial.

If the denominator polynomial is extracted and set equal to zero, that is: $D(s) = 0$, the resulting equation is called **the system's characteristic equation**, since it can be shown to characterize the system's dynamics. Its roots are the **poles** of $H(s)$. The roots of the numerator polynomial are the **zeros** of the transfer function. Both the poles and zeros of $H(s)$ can be complex values, $s = \sigma + j\omega$.

Example 2.6.3 Consider a system described by the transfer function

$$H(s) = \frac{s + 2}{s^3 + s^2 + s + 1}$$

It is a third order system because the degree of the denominator polynomial is 3. The system is proper since the degree of the denominator is greater than the degree of the numerator. The characteristic polynomial is:

$$D(s) = s^3 + s^2 + s + 1$$

The system's characteristic equation is:

$$s^3 + s^2 + s + 1 = 0$$

and the roots of the characteristic equation are the system poles, that result from:

$$s^3 + s^2 + s + 1 = (s + 1)(s^2 + 1) = 0, \text{ and the poles are: } p_1 = -1, p_{2,3} = \pm j$$

The numerator polynomial has one root, that is the zero of the system:

$$s + 2 = 0, \text{ and the zero is: } z_1 = -2$$

Example 2.6.4 Consider a system described by the transfer function

$$H(s) = \frac{2}{s^3(s + 3)}$$

It is a fourth order system because the degree of the denominator polynomial is 4. The system is proper since the degree of the denominator is greater than the degree of the numerator. The characteristic polynomial is:

$$D(s) = s^3(s + 3) = s^4 + 3s^3$$

The system's characteristic equation is:

$$s^4 + 3s^3 = 0$$

and the roots of the characteristic equation are the system poles, that result from:

$$s^4 + 3s^3 = s^3(s + 3) = 0, \text{ and the poles are: } p_1 = p_2 = p_3 = 0, p_4 = -3$$

The numerator is a zero-order polynomial, i.e. system has no zeros.

2.6.2 System response

The transfer function can be easily used for the calculation of system response to an input signal. Given the input signal $r(t)$ and the transfer function model of a system $H(s)$, we are interested in finding the system behavior, that is the output signal $y(t)$ (see Figure 2.10). From the definition of the transfer function:

$$Y(s) = H(s) \cdot R(s) \quad (2.25)$$

By applying the inverse Laplace transform we obtain:

$$y(t) = \mathcal{L}^{-1}[H(s) \cdot R(s)]. \quad (2.26)$$

Example 2.6.5 Spring-mass-damper response. Consider the spring-mass-damper system from the previous example and Figure 2.2. The transfer function of this system is given by:

$$H(s) = \frac{Y(s)}{R(s)} = \frac{1}{Ms^2 + fs + k}.$$

We shall obtain the system response or output variable, when the input $r(t)$ is a Dirac impulse and the system constants M, f, k are:

1. $M = 1, f = 3, k = 2$
2. $M = 1, f = 1, k = 2$ (smaller friction coefficient)

If $r(t)$ is a unit impulse, the Laplace transform will be $R(s) = 1$ and the output variable $y(t)$ is:

$$y(t) = \mathcal{L}^{-1}[H(s) \cdot R(s)] = \mathcal{L}^{-1}[H(s) \cdot 1] = \mathcal{L}^{-1}[H(s)]$$

1. In the first case the transfer function has two real poles

$$Y(s) = \frac{1}{(s+1)(s+2)}$$

Expanding $Y(s)$ in a partial fraction expansion, we obtain

$$Y(s) = \frac{k_1}{s+1} + \frac{k_2}{s+2}$$

where k_1 and k_2 are the coefficients of the expansion, or the residues, and are evaluated by multiplying through by the denominator factor corresponding to k_i and setting s equal to the root.

$$k_1 = \frac{1}{s+2} \Big|_{s=-1} = 1$$

$$k_2 = \frac{1}{s+1} \Big|_{s=-2} = -1$$

The inverse Laplace transform of $Y(s)$ is then

$$y(t) = \mathcal{L}^{-1} \left[\frac{1}{s+1} \right] - \mathcal{L}^{-1} \left[\frac{1}{s+2} \right]$$

Using a Laplace transforms table, we find that

$$y(t) = e^{-t} - e^{-2t}$$

The response $y(t)$ is called overdamped (no oscillations) and is shown in Figure 2.12 (left).

2. In this case $H(s)$ can be written as:

$$H(s) = \frac{1}{s^2 + s + 2}$$

and has the complex poles:

$$r_{1,2} = \frac{-1 \pm \sqrt{7}j}{2}$$

The system response will be called underdamped (damped oscillations) as the poles of the transfer function are complex and, as in the previous case, is computed from:

$$y(t) = \mathcal{L}^{-1}[H(s)]$$

Using a table with Laplace transform properties, and writing $Y(s)$ as

$$Y(s) = \frac{2}{\sqrt{7}} \frac{\frac{\sqrt{7}}{2}}{(s + \frac{1}{2})^2 + (\frac{\sqrt{7}}{2})^2}$$

the system response is:

$$y(t) = \frac{2}{\sqrt{7}} e^{-t/2} \sin\left(\frac{\sqrt{7}}{2}t\right)$$

The underdamped response is shown in Figure 2.12 (right).

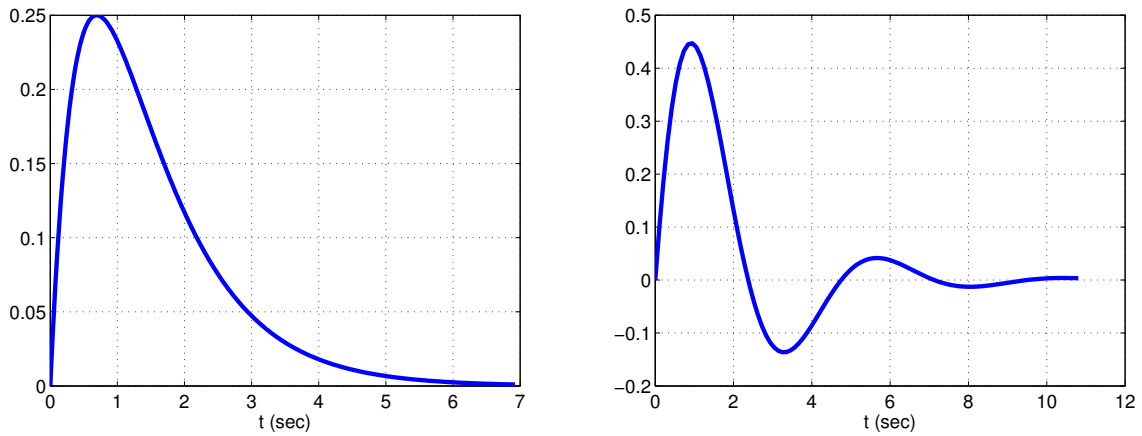


Figure 2.12: Spring-mass-damper response. Overdamped case (left). Underdamped case (right)

2.6.3 The transfer matrix

Because of the great number of systems with multiple inputs and outputs (MIMO systems), calculus techniques specific for this kind of systems were developed. In case of linear systems the mathematical techniques use transfer matrices instead of transfer functions.

Figure 2.13 illustrates a system with m inputs and n outputs.

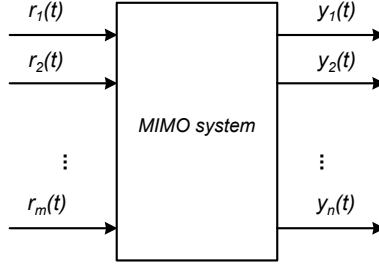


Figure 2.13: MIMO system

If the signal transfer is unidirectional and each input influences each output, as shown in Figure 2.14, we can write the following equation system:

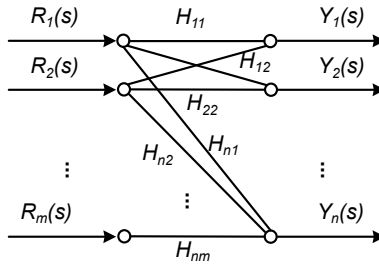


Figure 2.14: Transfer functions between inputs and outputs

$$\begin{aligned} Y_1(s) &= H_{11}R_1(s) + H_{12}R_2(s) + \dots + H_{1m}R_m(s) \\ Y_2(s) &= H_{21}R_1(s) + H_{22}R_2(s) + \dots + H_{2m}R_m(s) \\ &\vdots \\ Y_n(s) &= H_{n1}R_1(s) + H_{n2}R_2(s) + \dots + H_{nm}R_m(s) \end{aligned}$$

where $R(s)$ and $Y(s)$ are the Laplace transforms of the inputs $r(t)$ and outputs $y(t)$, and

$$H_{jk} = \frac{Y_j(s)}{R_k(s)}$$

is the transfer function from the input k to the output j .

Thus, a multiple linear system is composed by $n \times m$ elements having one input and one output. The equation system can be written in a matrix form:

$$\mathbf{Y}(s) = \mathbf{H}(s) \cdot \mathbf{R}(s)$$

where

$$\mathbf{R}(s) = [R_1(s) \ R_2(s) \ \dots \ R_m(s)]^T$$

is the $(m \times 1)$ input vector,

$$\mathbf{Y}(s) = [Y_1(s) \ Y_2(s) \ \dots \ Y_n(s)]^T$$

is the $(n \times 1)$ output vector , and

$$\mathbf{H} = \begin{bmatrix} H_{11}(s) & H_{12}(s) & \dots & H_{1m}(s) \\ H_{21}(s) & H_{22}(s) & \dots & H_{2m}(s) \\ \dots & \dots & \dots & \dots \\ H_{n1}(s) & H_{n2}(s) & \dots & H_{nm}(s) \end{bmatrix}$$

is called the **transfer matrix**.

Example 2.6.6 Consider a system with one input $r(t)$ and two outputs $y_1(t)$ and $y_2(t)$. If a linear relationship can be determined between the input and each output, then the system can be described by a transfer matrix having the size 2×1 . As an example (Figure 2.15) consider the transfer functions between the input and each output as follows:

$$H_{11}(s) = \frac{s}{s+1}, \quad H_{21}(s) = \frac{1}{s+2}$$

The relation between the input and outputs written in a matrix form is then:

$$\mathbf{Y}(s) = \begin{bmatrix} Y_1(s) \\ Y_2(s) \end{bmatrix} = \begin{bmatrix} H_{11}(s) \\ H_{21}(s) \end{bmatrix} U(s)$$

where the transfer matrix is:

$$\mathbf{H}(s) = \begin{bmatrix} \frac{s}{s+1} \\ \frac{1}{s+2} \end{bmatrix}$$

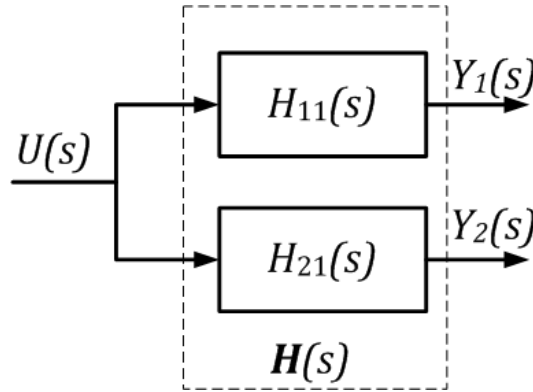


Figure 2.15: Transfer matrix example

2.7 Block Diagram Models

Since control systems are concerned with the control of specific signals, the interrelationship of the control variables to the controlling variables is required. This relationship is typically represented by the transfer function of the subsystem relating the input and output variables. Therefore one can correctly assume that the transfer function is an important relation for control engineering.

The importance of the cause-effect relationship of the transfer function is evidenced by the facility to represent the relationship of the system variables by schematic means. The **block diagram** representation of the system relationship is prevalent in control system engineering. Block diagrams consist of *unidirectional*, operational blocks that represent the transfer functions of the variables of interest, (Dorf and Bishop, 2008).

The block diagram representation of a given system can often be reduced, using block diagram reduction techniques. The resulting diagram obeys the law of superposition because of the linear behavior of each subsystem.

In control engineering, the block diagram is a primary tool that together with transfer functions can be used to describe cause-and-effect relationships throughout a dynamic system. The manipulation of block diagrams adheres to a mathematical system of rules often known as block diagram algebra (Mei, 2002).

In general, the block diagram of a linear time invariant system consists of four components, namely signal, block (with transfer function), summing point and pickoff point as shown in Figure 2.16, (Mei, 2002).

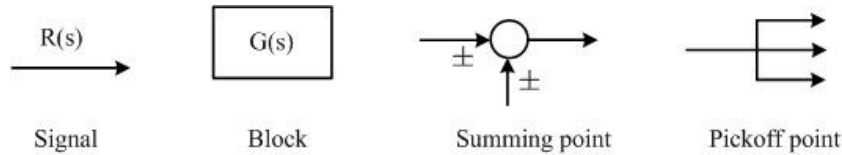


Figure 2.16: The components of a block diagram

2.7.1 Basic connections

In Figure 2.17 are shown the three basic connections: series (cascade), parallel and feedback. The equivalent transfer function from the input $R(s)$ to the output $Y(s)$ for each of these cases can be determined as follows:

1. Series connection.

$$H_1(s) = \frac{Y_1(s)}{R_1(s)}; H_2(s) = \frac{Y_2(s)}{R_2(s)}; Y_1(s) = R_2(s)$$

The equivalent transfer function can be calculated from:

$$Y(s) = Y_2(s) = H_2(s)R_2(s) = H_2(s)Y_1(s) = H_2(s)H_1(s)R_1(s) = \underbrace{H_2(s)H_1(s)}_{H(s)} R(s)$$

$$H(s) = \frac{Y(s)}{R(s)} \Rightarrow H(s) = H_1(s)H_2(s)$$

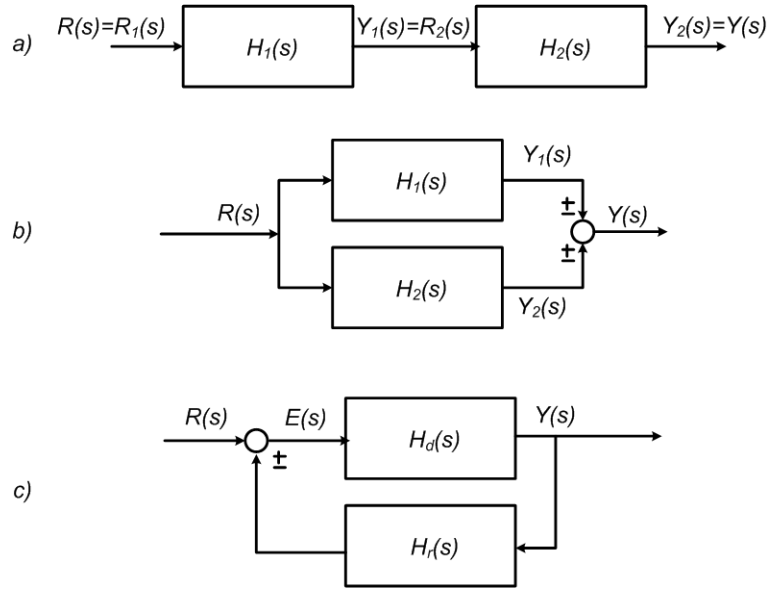


Figure 2.17: Three basic connections. a) series, b) parallel, c) feedback

For n systems connected in cascade the equivalent transfer function will be:

$$H(s) = \prod_{j=1}^n H_j(s)$$

2. Parallel connection.

$$H_1(s) = \frac{Y_1(s)}{R(s)}; H_2(s) = \frac{Y_2(s)}{R(s)}; Y(s) = \pm Y_1(s) \pm Y_2(s)$$

The equivalent transfer function is:

$$H(s) = \frac{Y(s)}{R(s)} = \pm H_1(s) \pm H_2(s)$$

For n systems connected in parallel the equivalent transfer function is:

$$H(s) = \sum_{j=1}^n (\pm H_j(s))$$

3. Feedback connection. The error signal $E(s)$ can be calculated from:

$$E(s) = R(s) \pm H_r(s) \cdot Y(s)$$

The output is:

$$Y(s) = H_d(s) \cdot E(s)$$

If $E(s)$ is eliminated between the previous two relations, the overall transfer function will be:

$$H(s) = \frac{Y(s)}{R(s)} = \frac{H_d(s)}{1 \mp H_d(s) \cdot H_r(s)}$$

Note that in the relation above the sign is $+$ for negative feedback and $-$ for positive feedback.

Example 2.7.1 Consider a system represented by the block diagram shown in Figure 2.18. Determine the closed-loop transfer function $H(s) = Y(s)/R(s)$.

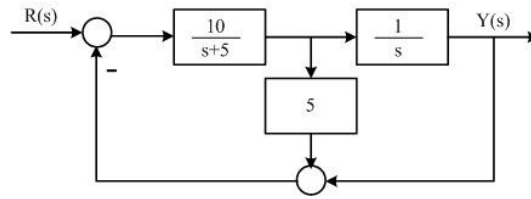


Figure 2.18: Block diagram example exercise.

The block diagram will be transformed so the basic rules of block diagram reduction can be applied. First, the summing points encircled in Figure 2.19 (left) will be considered. Note that the signal Z can be calculated as:

$$Z = R - (X + Y) = R - X - Y \quad (2.27)$$

and the two summing points can be reduced to a single one, as shown in Figure 2.19 (right)

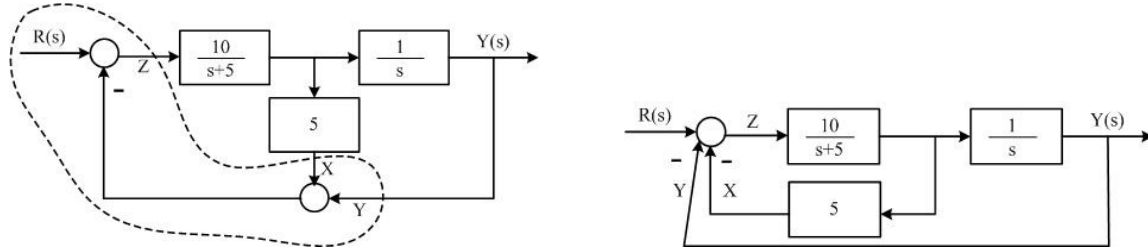


Figure 2.19: Block diagram transformation.

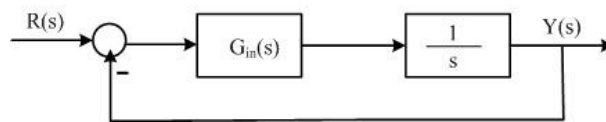


Figure 2.20: Block diagram transformation.

From now on, the basic rules apply. The inner loop is reduced first to one block and the equivalent transfer function is:

$$G_{in}(s) = \frac{\frac{10}{s+5}}{1 + \frac{10}{s+5} \cdot 5} = \frac{10}{s+55} \quad (2.28)$$

This resulting block, is in a series connection with $1/s$ and the equivalent of these two is the product G_{in}/s . The new block diagram is presented in Figure 2.20.

Note that the feedback path has a transfer function equal to 1, therefore it is a unity feedback loop. The equivalent transfer function from the input $R(s)$ to the output $Y(s)$ is calculated as follows:

$$H(s) = \frac{Y(s)}{R(s)} = \frac{G_{in}(s) \cdot \frac{1}{s}}{1 + G_{in}(s) \cdot \frac{1}{s} \cdot 1} = \frac{\frac{10}{s(s+55)}}{1 + \frac{10}{s(s+55)}} = \frac{10}{s^2 + 55s + 10} \quad (2.29)$$

Diagram transformations

1. Point behind a block - Figure 2.21

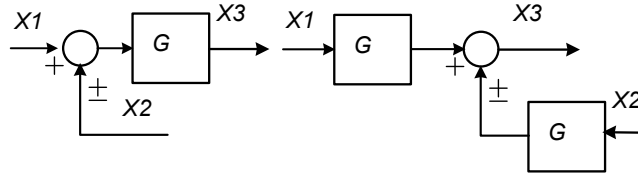


Figure 2.21: Point behind a block

2. Moving a pickoff point ahead of a block - Figure 2.22

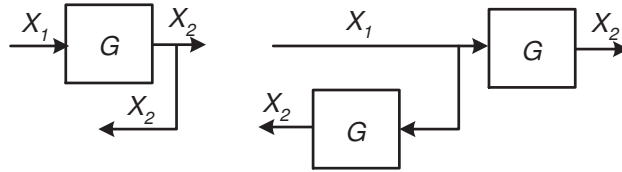


Figure 2.22: Moving a pickoff point ahead of a block

3. Moving a pickoff point behind a block - Figure 2.23

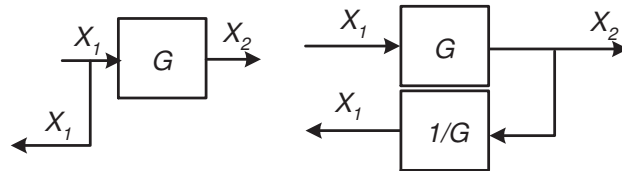


Figure 2.23: Moving a pickoff point behind a block

4. Moving a summing point ahead of a block - Figure 2.24

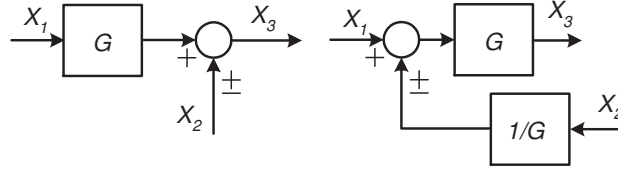


Figure 2.24: Moving a summing point ahead of a block

2.7.2 The overlap of signals

If a system has more than one input, then the output can be considered a result of all effects of these signals.

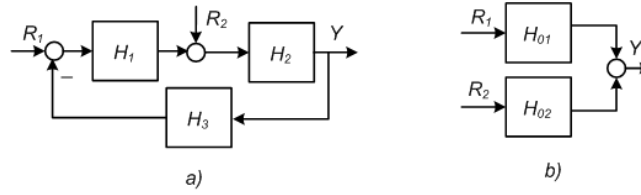


Figure 2.25: Two-input system

Consider the system shown in Figure 2.25 a). Since all linear systems satisfy the principle of superposition, the total output of the system is the algebraic sum (superposition) of the outputs due to each of the inputs. In a general case, all inputs except one are set equal to zero, and then the output due to the non-zero input is determined.

For the system considered above, a transfer function relating each input and the output can be calculated, when the other input is considered zero (Figure 2.25 b)).

$$Y(s) = R_1(s) \cdot H_{01}(s)|_{R_2(s)=0} + R_2(s) \cdot H_{02}(s)|_{R_1(s)=0}$$

The block diagram from Figure 2.25 will be re-arranged for each non-zero input.

When $R_2 = 0$ (see Figure 2.26) the equivalent transfer function between $R_1(s)$ and its effect, or output, $Y_1(s)$ is:

$$H_{01}(s) = \frac{H_1(s)H_2(s)}{1 + H_1(s)H_2(s)H_3(s)} \text{ and the output: } Y_1(s) = H_{01}(s)R_1(s)$$

When $R_1 = 0$ (see Figure 2.27) the equivalent transfer function between $R_2(s)$ and the output $Y_2(s)$ is:

$$H_{02}(s) = \frac{H_2(s)}{1 + H_1(s)H_2(s)H_3(s)} \text{ and the output: } Y_2(s) = H_{02}(s)R_2(s)$$

Thus, the output of the system shown in Figure 2.25 is given by:

$$Y(s) = \frac{H_1H_2}{1 + H_1H_2H_3} \cdot R_1(s) + \frac{H_2}{1 + H_1H_2H_3} \cdot R_2(s)$$

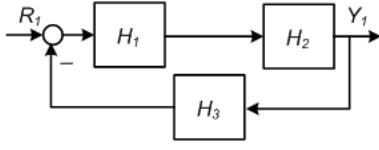


Figure 2.26: Block diagram for $R_2 = 0$

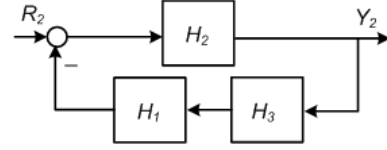


Figure 2.27: Block diagram for $R_1 = 0$

2.7.3 The algebra of MIMO systems

In case of connecting linear MIMO systems, we can use transfer matrices instead of transfer functions, obeying the rules of matrix calculus.

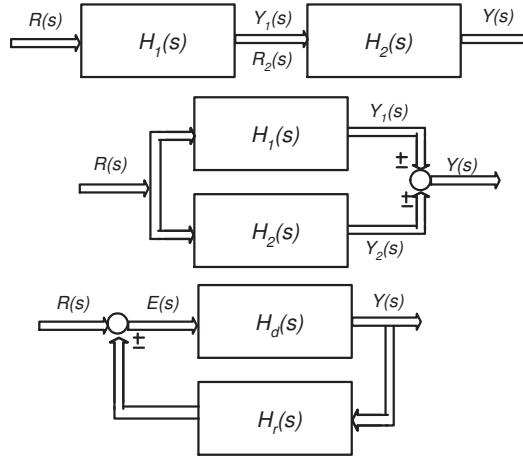


Figure 2.28: MIMO system

For example consider the three basic connections shown in Figure 2.28.

1. Series connection.

$$\mathbf{Y}_1 = \mathbf{H}_1 \cdot \mathbf{R}; \quad \mathbf{Y} = \mathbf{H}_2 \cdot \mathbf{R}_2; \quad \mathbf{R}_2 = \mathbf{Y}_1; \quad \Rightarrow \quad \mathbf{Y} = \mathbf{H}_2 \cdot \mathbf{H}_1 \cdot \mathbf{R}$$

The equivalent transfer matrix will be:

$$\mathbf{H} = \mathbf{H}_2 \cdot \mathbf{H}_1$$

For n linear systems connected in series, the equivalent transfer matrix is:

$$\mathbf{H} = \prod_{j=1}^n \mathbf{H}_j$$

2. Parallel connection.

$$\mathbf{Y}_1 = \mathbf{H}_1 \cdot \mathbf{R}; \quad \mathbf{Y}_2 = \mathbf{H}_2 \cdot \mathbf{R}; \quad \mathbf{Y} = \pm \mathbf{Y}_1 \pm \mathbf{Y}_2$$

The equivalent transfer matrix will be:

$$\mathbf{H} = \pm \mathbf{H}_1 \pm \mathbf{H}_2$$

3. Feedback connection.

$$\mathbf{E} = \mathbf{R} \pm \mathbf{H}_r \cdot \mathbf{Y}; \quad \mathbf{Y} = \mathbf{H}_d \cdot \mathbf{E}; \quad \mathbf{Y} = \mathbf{H}_d \cdot (\mathbf{R} \pm \mathbf{H}_r \cdot \mathbf{Y})$$

$$(\mathbf{1} \mp \mathbf{H}_d \cdot \mathbf{H}_r) \cdot \mathbf{Y} = \mathbf{H}_d \cdot \mathbf{R}; \quad \mathbf{Y} = (\mathbf{1} \mp \mathbf{H}_d \cdot \mathbf{H}_r)^{-1} \cdot \mathbf{H}_d \cdot \mathbf{R}$$

The equivalent transfer matrix will be:

$$\mathbf{H} = (\mathbf{1} \mp \mathbf{H}_d \cdot \mathbf{H}_r)^{-1} \cdot \mathbf{H}_d$$

Bibliography

- Dorf, R. C. and Bishop, R. H. (2008). *Modern Control Systems*. Pearson Prentice Hall, 11th edition.
- Dutton, K., Thompson, S., and Barraclough, B. (1997). *The Art of Control Engineering*. Addison-Wesley.
- Mei, C. (2002). On teaching the simplification of block diagrams. *International Journal of Engineering Education*, 18(6):697–703.
- Xie, Y. (2009). Maglev demonstration. web: web.mit.edu/2.737/www/MagLev/.